

Frobenius Groups

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Introduction

The representation theory of finite groups, discovered by Frobenius toward the end of the 19th century, consists in analyzing the properties and the structure of a finite group $(G, \cdot)$ by making it act linearly on a vector space. In this memoir (which was originally written as a TIPE, a supervised personal initiative project, for the École Normale Supérieure entrance examination), we use representation theory to prove a theorem about Frobenius groups. In the appendix, we collect several prerequisites that we will use without further comment in this text.

Note that the original document was in French. It was automatically translated into English using an LLM, please report any errors.

1 The Frobenius theorem

Definition 1. Let $(G, \cdot)$ be a finite group with identity e and H a proper subgroup of G (i.e. $H \neq \{e\}, H \neq G$). We say that G is a Frobenius group, that H is a Frobenius complement, and that (G, H) is a Frobenius pair if

$$\forall g \in G \setminus H, gHg^{-1} \cap H = \{e\}$$

This condition is equivalent to $\forall (x, y) \in G^2, y^{-1}x \notin H \Rightarrow xHx^{-1} \cap yHy^{-1} = \{e\}$. Intuitively, this means that the conjugates of H cover G "as efficiently as possible".

Proposition 1. If (G, H) is a Frobenius pair, $|\bigcup_{g \in G} gHg^{-1}| = |G| + 1 - \frac{|G|}{|H|}$.

Proof. $|\bigcup_{g \in G} gHg^{-1}| = |\{e\} \sqcup \bigsqcup_{gH \in G/H} (gHg^{-1} \setminus \{e\})| = 1 + |G/H|(|H| - 1)$

□

It can be shown that for an arbitrary subgroup H the equality above is an inequality $\leq$. This also shows that the conjugates of a proper subgroup never cover G (recall that G is finite, the triangular matrices provide a counterexample in the infinite group $GL_n(\mathbb{C})$).

Example 1. Let G be a finite group acting transitively on a set X , such that for $g \in G \setminus \{e\}$, g fixes at most one point of X . Then, if $\alpha \in X$ is such that $\text{Stab}(\alpha) \neq \{e\}$, $(G, \text{Stab}(\alpha))$ is a Frobenius pair.

Proof. We have $\text{Stab}(\alpha) \neq G$ (by transitivity). If $g \in G \setminus \text{Stab}(\alpha)$, then $g \cdot \alpha \neq \alpha$ so $g \cdot \alpha$ and α are fixed simultaneously only by e , hence $\text{Stab}(g \cdot \alpha) \cap \text{Stab}(\alpha) = \{e\}$ that is, $g\text{Stab}(\alpha)g^{-1} \cap \text{Stab}(\alpha) = \{e\}$. □

In fact, one can show that every Frobenius group is of this form. Here are some examples:

Example 2. • If $\mathbb{K}$ is a finite field and $G = \text{Aff}(1, \mathbb{K}) = \{\tau_{a,b} : x \in \mathbb{K} \mapsto ax + b, (a, b) \in \mathbb{K}^* \times \mathbb{K}\}$ is the group of affine bijections of $\mathbb{K}$ onto itself and $H(\mathbb{K}) = \{\tau_{a,0} : x \in \mathbb{K} \mapsto ax, a \in \mathbb{K}^*\}$ the subgroup of elements fixing 0, then (G, H) is a Frobenius pair ($\tau_{a,b}$ has exactly one fixed point, namely $\frac{b}{1-a}$ for $a \neq 1$, has no fixed point of $\mathbb{K}$ for $a = 1$ and $b \neq 0$, fixes all of $\mathbb{K}$ for $a = 1$ and $b = 0$ that is, for $\tau_{a,b} = id$; and finally $H(\mathbb{K}) = \text{Stab}(0)$)

- If $G = \mathcal{A}_4$ (alternating group) and γ a 3-cycle, then $(G, \langle \gamma \rangle)$ is a Frobenius pair
- If $n \in \mathbb{N}^*$ is odd, D_n the dihedral group (affine transformations of the plane preserving the regular n -gon centred at the origin), s an reflection in D_n , then $(D_n, \langle s \rangle)$ is a Frobenius pair.

Theorem 1 (Frobenius). *Let (G, H) be a Frobenius pair. We set*

$$N = \left(G \setminus \bigcup_{g \in G} gHg^{-1} \right) \cup \{e\}$$

called the Frobenius kernel. Then N is a normal subgroup of G and $N \cap H = \{e\}$ and $G = NH$ (we say that G is the semidirect product of N by H).

$N \cap H = \{e\}$ is obvious, moreover, by proposition 1, $|N| = |G| - |\bigcup_{g \in G} gHg^{-1}| + 1 = |G|/|H|$. Assuming that N is a subgroup of G , $\varphi : (n, h) \in N \times H \mapsto nh$ is injective (if $nh = n'h'$ then $n'^{-1}n = h'h^{-1} \in N \cap H = \{e\}$ so $(n, h) = (n', h')$) and is therefore surjective by cardinality; hence $G = NH$. Moreover if $g \in G$, $gNg^{-1} \subseteq N$ (otherwise an element of $N \setminus \{e\}$ would be conjugate to an element of H contradicting the definition of N) so N is normal. The difficult part of the theorem is proving that N is a subgroup. This will be proved in section 5.

Example 3. We illustrate this using the pairs of example 2. In the "group action" formulation of example 1, the Frobenius kernel is the set of elements of G acting without a fixed point, together with the identity e . Thus,

- The kernel of $(\mathcal{A}_4, < \gamma >)$ is the Klein group $V = \{id, (1\ 2)(3\ 4), (1\ 3)(2\ 4), (1\ 4)(2\ 3)\}$
- The kernel of $(\text{Aff}(1, \mathbb{K}), H(\mathbb{K}))$ is the subgroup $\{\tau_{1,b}, b \in \mathbb{K}\}$ of translations of $\text{Aff}(1, \mathbb{K})$
- The Frobenius kernel of D_n is the group of rotations centered at the origin by an angle in $\frac{2\pi}{n}\mathbb{Z}$ (naturally isomorphic to $\mathbb{U}_n$)

2 Representations of a finite group

To prove Frobenius' theorem, we will seek to express N as the kernel of a representation.

Definition 2. *Let $(G, \cdot)$ be a finite group. A representation of G is a pair (ρ, V) where V is a finite-dimensional $\mathbb{C}$ -vector space and ρ is a homomorphism from G to $\text{GL}(V)$. We will sometimes write only ρ . The degree of ρ is the integer $\dim(V)$.*

Thus we view the elements of G as linear transformations on V .

Example 4. • Representations of degree 1 can be identified with the homomorphisms from G to $\mathbb{C}^*$. In particular, we call the representation corresponding to the trivial homomorphism from G to $\mathbb{C}^*$ the trivial representation.

- A more substantial example: if $(f_g)_{g \in G}$ is a basis indexed by G of a vector space V of dimension $|G|$, define, for $g \in G$, $\rho(g)$ to be the unique element of $\text{GL}(V)$ satisfying $\forall h \in G, \rho(g)(f_h) = f_{gh}$. This defines a representation of G , called the regular representation. We also say that it is "faithful" because $\ker(\rho) = \{e\}$.
- If (ρ, V) and (ρ', V') are two representations of G , then, for $g \in G$, we define a representation $(\widetilde{(\rho, \rho')}, \mathcal{L}(V, V'))$ by setting: $\forall u \in \mathcal{L}(V, V'), \widetilde{(\rho, \rho')}(g)(u) = \rho'(g) \circ u \circ \rho(g^{-1})$,
- Let (ρ, V) and (ρ', V') be as above. We then define a representation $(\rho \oplus \rho', V \times V')$ by setting $\forall (g, v, v') \in G \times V \times V', (\rho \oplus \rho')(g)((v, v')) = (\rho(g)(v), \rho'(g)(v'))$. This construction also applies for subspaces in a direct sum on which ρ and ρ' are defined: in a basis adapted to this decomposition, the matrices of the $(\rho \oplus \rho')(g)$ will consist of 2 square diagonal blocks of sizes $\dim(V)$ and $\dim(V')$.

Definition 3. *If (ρ, V) and (ρ', V') are two representations of G , we say that ρ is equivalent to ρ' , and we write $\rho \sim \rho'$ if $\dim(V) = \dim(V')$ and if there exist bases e, e' of V and V' such that*

$$\forall g \in G, \text{Mat}_e(\rho(g)) = \text{Mat}_{e'}(\rho'(g))$$

i.e. if there exists $u \in \mathcal{L}(V, V')$ an isomorphism such that $\forall g \in G, u \circ \rho(g) = \rho'(g) \circ u$. Moreover, for arbitrary ρ, ρ' , we write $\text{Hom}(\rho, \rho') = \{u \in \mathcal{L}(V, V'), \forall g \in G, u \circ \rho(g) = \rho'(g) \circ u\}$ which is clearly a subspace of $\mathcal{L}(V, V')$.

Definition 4. *A representation ρ of G is said to be irreducible if the only subspaces stable under all the $\rho(g)$ for $g \in G$ are $\{0\}$ and V .*

Such representations cannot be decomposed into proper subrepresentations, this is the case, for example, for the representations of degree 1. Indeed, we have the following proposition:

Proposition 2 (Maschke). *Let (ρ, V) be a representation of G and U a subspace of V stable under all the $\rho(g)$ for $g \in G$. There exists a complement S of U which is itself stable under all the $\rho(g)$.*

Proof. It is enough to construct a projector with image U that commutes with all the $\rho(g)$, its kernel will then be a desired complement. Let p be an arbitrary projector with image U . We modify it by averaging over G :

$$q = \frac{1}{|G|} \sum_{g \in G} \rho(g) \circ p \circ \rho(g^{-1})$$

This is indeed an endomorphism with image included in U such that $q|_U = id_U$ (because the $\rho(g)$ stabilize U) and it commutes with the $\rho(g)$ ($\rho(g) \circ q \circ \rho(g)^{-1} = q$ because $h \in G \mapsto gh \in G$ is bijective). $\square$

By induction on $\dim(V)$, we obtain the following co-reduction result:

Proposition 3. *If (ρ, V) is a representation, there exist $k \in \mathbb{N}^*$ and $V_1, V_2, \dots, V_k$, subspaces of V , such that $V = \bigoplus_{i=1}^k V_i$, and irreducible representations $(\rho_1, V_1), \dots, (\rho_k, V_k)$ such that*

$$\rho = \rho_1 \oplus \rho_2 \oplus \dots \oplus \rho_k$$

Finally, we state one last proposition that will be very useful:

Proposition 4 (Schur). *Let (ρ, V) and (ρ', V') be two irreducible representations of G . Then $\dim \text{Hom}(\rho, \rho') = \mathbb{1}_{\rho \sim \rho'}$*

Proof. • Suppose $\rho \sim \rho'$. After transporting ρ' by the isomorphism given by the equivalence, suppose $V = V'$ and $\rho = \rho'$. Let $u \in \text{Hom}(\rho, \rho)$. Since $\mathbb{C}$ is algebraically closed, u has an eigenvalue λ . Then for all g in G , u and $\rho(g)$ commute so $E_\lambda(u) = \ker(u - \lambda id)$ is stable under $\rho(g)$. By irreducibility of ρ , since $\ker(u - \lambda id)$ is nonzero, it is V and $u = \lambda id$. Conversely, the maps λid , with $\lambda \in \mathbb{C}$, are in $\text{Hom}(\rho, \rho)$. Finally $\text{Hom}(\rho, \rho) = \mathbb{C}id$ is of dimension 1.

• Suppose $\rho \not\sim \rho'$. Let $u \in \text{Hom}(\rho, \rho')$. If $g \in G$, $\rho'(g) \circ u = u \circ \rho(g)$ so $\text{im}(u)$ is stable under $\rho'(g)$. By irreducibility, $\text{im}(u) = V'$ or $\{0\}$. Likewise, $\ker(u) = V$ or $\{0\}$. So since u is not an isomorphism, $\ker(u) = V$ and $\text{im}(u) = \{0\}$ and $u = 0$ which completes the proof. $\square$

3 Characters

A substantial part of the information contained in a representation is in fact stored in its character:

Definition 5. *Let (ρ, V) be a representation of G . The character of ρ is the function*

$$\begin{aligned} \chi_\rho : G &\longrightarrow \mathbb{C} \\ g &\longmapsto \text{Tr}(\rho(g)) \end{aligned}$$

A character is any function in $\mathbb{C}^G$ that arises as the character of a representation. A character is called irreducible if it comes from an irreducible representation

Example 5. • $\chi(e) = \text{Tr}(id)$ is the degree of the representation

- 2 equivalent representations have the same character (their matrices are the same in suitable bases)
- If $g, h \in G$, χ a character, $\chi(ghg^{-1}) = \text{Tr}(\rho(g)\rho(h)\rho(g)^{-1}) = \text{Tr}(\rho(h)) = \chi(h)$. We denote by $\text{Cent}(G)$ the subspace of central functions of $\mathbb{C}^G$, i.e. constant on the conjugacy classes of G (of dimension the number of such classes). Then $\chi \in \text{Cent}(G)$.
- We trivially have $\chi_{\rho_1 \oplus \rho_2} = \chi_{\rho_1} + \chi_{\rho_2}$
- For $g \in G$, $g^{|G|} = e$ so $X^{|G|} - 1 = \prod_{\omega \in \mathbb{U}_{|G|}} (X - \omega)$ annihilates $\rho(g)$ and is split with simple roots so $\rho(g)$ is diagonalizable with spectrum in $\mathbb{U}_{|G|}$. If $\{\lambda_1, \dots, \lambda_d\}$ is the spectrum of $\rho(g)$, $\{\lambda_1^{-1}, \dots, \lambda_d^{-1}\}$ is that of $\rho(g)^{-1} = \rho(g^{-1})$ hence, since the spectrum is in $\mathbb{U}$, $\chi(g^{-1}) = \lambda_1^{-1} + \dots + \lambda_d^{-1} = \overline{\lambda_1} + \dots + \overline{\lambda_d} = \overline{\chi(g)}$
- Using the notation of example 4, $\widetilde{\chi_{(\rho, \rho')}} = \overline{\chi_\rho} \chi_{\rho'}$ (this follows from the previous point and the definition of $\widetilde{(\rho, \rho')}$, of the fact that for $A, B \in \mathcal{M}_n(\mathbb{C})$, the linear map $M \in \mathcal{M}_n(\mathbb{C}) \mapsto AMB \in \mathcal{M}_n(\mathbb{C})$ has trace $\text{Tr}(A)\text{Tr}(B)$, because if $1 \leq i, j \leq n$, $AE_{i,j}B = \sum_{1 \leq k, l \leq n} A_{ki}B_{jl}E_{k,l}$ has coordinate $A_{ii}B_{jj}$ at $E_{i,j}$)

Definition/Proposition 1. *We equip $\mathbb{C}^G$ with the Hermitian inner product (cf. appendix) defined by:*

$$\forall (u, v) \in \mathbb{C}^G \times \mathbb{C}^G, \langle u, v \rangle = \frac{1}{|G|} \sum_{g \in G} \overline{u(g)} v(g)$$

To compute the Hermitian inner product of 2 characters, we again use the "averaging technique":

Lemma 1. *Let ρ be a representation, then $\frac{1}{|G|} \sum_{g \in G} \rho(g)$ is a projector onto $V^G := \bigcap_{g \in G} \ker(\rho(g) - id)$*

Proof. If $x \in V^G$, $\frac{1}{|G|} \sum_{g \in G} \rho(g)(x) = \frac{1}{|G|} \sum_{g \in G} x = x$ (1). Moreover, if $x \in V$ and $y = \frac{1}{|G|} \sum_{g \in G} \rho(g)(x)$, and $h \in G$, $\rho(h)(y) = \frac{1}{|G|} \sum_{g \in G} \rho(hg)(x) = y$ (because $g \mapsto hg$ is bijective) and so $y \in V^G$ which completes the proof (2). With (1) and (2), we obtain the desired result. $\square$

Proposition 5. If $(\rho, V), (\rho', V')$ are 2 representations of G , $\langle \chi_\rho, \chi_{\rho'} \rangle = \dim(\text{Hom}(\rho, \rho'))$

Proof. We set $\widetilde{(\rho, \rho')}$ as in example 4. We apply lemma 1 to this representation. On the one hand, $\cap_{g \in G} \ker(\widetilde{(\rho, \rho')}(g) - \text{id}_{\mathcal{L}(V, V')}) = \text{Hom}(\rho, \rho')$, so $\text{Tr}(\frac{1}{|G|} \sum_{g \in G} \widetilde{(\rho, \rho')}(g)) = \dim(\text{Hom}(\rho, \rho'))$ (by diagonalizing a projector, we see that its trace is its rank). On the other hand, by linearity and example 5, $\text{Tr}(\frac{1}{|G|} \sum_{g \in G} \widetilde{(\rho, \rho')}(g)) = \frac{1}{|G|} \sum_{g \in G} \chi_{\widetilde{(\rho, \rho')}}(g) = \frac{1}{|G|} \sum_{g \in G} \overline{\chi_\rho(g)} \chi_{\rho'}(g) = \langle \chi_\rho, \chi_{\rho'} \rangle$ $\square$

Combining this with Schur's lemma (proposition 4),

Corollary 1. If $(\rho, V), (\rho', V')$ are 2 irreducible representations of G , $\langle \chi_\rho, \chi_{\rho'} \rangle = \mathbb{1}_{\rho \sim \rho'}$. Thus the family of irreducible characters is orthonormal and is therefore finite ($\mathbb{C}^G$ is finite-dimensional). Moreover, 2 non-equivalent irreducible representations have distinct characters: therefore, there are as many irreducible representations up to equivalence as there are irreducible characters.

Let $\rho_1, \dots, \rho_r$ be a complete set of irreducible representations, and $\chi_1, \dots, \chi_r$ their associated characters, by combining with proposition 3,

Corollary 2. If ρ is a representation, there exists $(n_1, \dots, n_r) \in \mathbb{N}^r$ such that $\rho \sim n_1 \rho_1 \oplus \dots \oplus n_r \rho_r$ (where $n_i \rho_i$ denotes $\underbrace{\rho_i \oplus \dots \oplus \rho_i}_{n_i \text{ times}}$). Then $\chi = \sum_{i=1}^r n_i \chi_i$

Proposition 6. If ρ and ρ' are 2 representations, $\rho \sim \rho' \iff \chi_\rho = \chi_{\rho'}$

Proof. The direct implication is clear. If $\chi_\rho = \chi_{\rho'} =: \chi$, writing $\rho \sim n_1 \rho_1 \oplus \dots \oplus n_r \rho_r$ and $\rho' \sim n'_1 \rho_1 \oplus \dots \oplus n'_r \rho_r$, by orthonormality $\langle \chi_i, \chi_\rho \rangle = \sum_{j=1}^r n_j \langle \chi_i, \chi_j \rangle = n_i$. Likewise, $\langle \chi_i, \chi_{\rho'} \rangle = n'_i$. Then $n_i = n'_i$ and $\rho \sim \rho'$. $\square$

In fact, one can prove a stronger statement than the orthonormality of the characters (we will not prove it here):

Proposition 7. The irreducible characters $\chi_1, \dots, \chi_r$ form an orthonormal basis of $\text{Cent}(G)$. Thus r is the number of conjugacy classes in G and if $f \in \text{Cent}(G)$,

$$f = \sum_{i=1}^r \langle \chi_i, f \rangle \chi_i$$

4 Induced representations

Let H be a subgroup of G . From a representation (ρ, V) of G , we can easily obtain the representation $(\rho|_H, V)$ of H , denoted $\rho|_H$. Conversely, the goal of this section is to give a natural way to extend a representation (ρ_0, W) of H to a representation (ρ, V) of G (where $W \subseteq V$). We write

$$G = \bigsqcup_{i=1}^l g_i H$$

the partition of G into left cosets modulo H (where the g_i are chosen representatives and $g_1 H = H$).

Definition 6. Let (ρ, V) be a representation of G , and let W be a subspace of V stable under the $\rho(h), h \in H$. Let $\rho_0 = \rho|_{H, W}$ be the restriction of ρ to H and to the subspace W . We say that ρ is induced from ρ_0 and we write $\rho = \text{Ind}_H^G(\rho_0)$ if

$$V = \bigoplus_{i=1}^l \rho(g_i)(W) \quad (*)$$

Note that under the hypotheses of definition 6, if $g \in G$ is written $g_i h$ where $h \in H$, $\rho(g)(W) = \rho(g_i h)(W) = \rho(g_i) \circ \rho(h)(W) = \rho(g_i)(W)$. So the subspaces $\rho(g_i)(W)$ do not depend on the choice of representatives in the left coset modulo H . We have the following proposition:

Proposition 8. If ρ' is a representation of H , there exists a representation ρ of G , unique up to equivalence, of G such that $\rho = \text{Ind}_H^G(\rho')$

We do not prove this: we merely observe that if (ρ_0, W) and the action of the $\rho(g_i)$ on W are fixed, the whole induced representation is determined. Indeed, it suffices to determine it on the $\rho(g_i)(W)$ (by direct sum) and if $g \in G$, $1 \leq i \leq r$, $w \in W$, we have $\rho(g)(\rho(g_i)(w)) = \rho(g_j)(\rho_0(h)(w))$ where $gg_i = g_j h$ (we take the representative of the class of gg_i modulo H). The following result explains the usefulness of induced representations:

Proposition 9. Let (ρ, V) and (ρ', V') be two representations of G and (ρ_0, W) a representation of H . We suppose $\rho = \text{Ind}_H^G(\rho_0)$ (in particular, $\rho_0 = \rho|_{H, W}$). Then $\text{Hom}(\rho, \rho') \simeq \text{Hom}(\rho_0, \rho'_H)$

Proof. $u \in \text{Hom}(\rho, \rho') \subseteq \mathcal{L}(V, V') \mapsto u|_W \in \text{Hom}(\rho_0, \rho'_H) \subseteq \mathcal{L}(W, V')$ is linear. To prove bijectivity, let us show that an element $v \in \text{Hom}(\rho_0, \rho'_H)$ extends on W into a unique element of $\text{Hom}(\rho, \rho')$: if u is such an extension, $\forall 1 \leq i \leq l, \forall w \in W, u(\rho(g_i)(w)) = \rho'(g_i)(u(w)) = \rho'(g_i)(v(w))$ which defines u on the whole of V by the direct sum $(*)$. Conversely, one readily checks that such a u works. $\square$

Translating proposition 9 into an equality of dimensions (by proposition 5),

Corollary 3. Using the notation of proposition 9, writing $\text{Ind}_H^G(\chi_{\rho_0}) = \chi_\rho$, and $\langle \cdot, \cdot \rangle_G$ (resp. $\langle \cdot, \cdot \rangle_H$) the standard inner product on $\text{Cent}(G)$ (resp. $\text{Cent}(H)$), we have

$$\langle \text{Ind}_H^G(\chi_{\rho_0}), \chi_{\rho'} \rangle_G = \langle \chi_{\rho_0}, (\chi'_{\rho'})|_H \rangle_H$$

By proposition 6, we can speak of the "induced character" of χ_{ρ_0} for $\text{Ind}_H^G(\chi_{\rho_0})$.

Proposition 10. If χ is a character of H ,

$$\forall g \in G, \text{Ind}_H^G(\chi)(g) = \frac{1}{|H|} \sum_{k \in G, kgk^{-1} \in H} \chi(kgk^{-1})$$

Proof. We write (ρ, V) and (ρ', W) such that $\chi = \chi_{\rho'}$ and $\rho = \text{Ind}_H^G(\rho')$. Let $g \in G$. $\rho(g)$ permutes the $\rho(g_i)(W)$ hence, by looking at the matrices of the $\rho(g)$ in a basis adapted to $V = \bigoplus_{i=1}^l \rho(g_i)(W)$, to compute the trace of $\rho(g)$, it suffices to consider those i such that $\rho(g)(\rho(g_i)(W)) = \rho(g_i)(W)$ i.e. such that $\rho(g_i^{-1}gg_i)(W) = W$ i.e. $g_i^{-1}gg_i \in H$. Thus, by invariance of the trace under change of basis (we transport $\rho(g_i)(W)$ onto W by the isomorphism $\rho(g_i)$) $\chi(g) = \sum_{1 \leq i \leq l, g_i^{-1}gg_i \in H} \text{Tr}(\rho(g)|_{\rho(g_i)(W)}) = \sum_{1 \leq i \leq l, g_i^{-1}gg_i \in H} \text{Tr}(\rho(g_i^{-1}gg_i)|_W) = \sum_{1 \leq i \leq l, g_i^{-1}gg_i \in H} \chi(g_i^{-1}gg_i)$ which completes the proof because χ is central on H and the right cosets modulo H have cardinality $|H|$. $\square$

This formula allows us to define $\text{Ind}_H^G(f)$ for any central function f (replacing χ by f in the formula). Since the characters linearly span the central functions, we deduce from corollary 3:

Proposition 11. If $f_1 \in \text{Cent}(H)$ and $f_2 \in \text{Cent}(G)$, we have

$$\langle \text{Ind}_H^G(f_1), f_2 \rangle_G = \langle f_1, f_2|_H \rangle_H$$

5 Proof of the Frobenius theorem

We now prove of theorem 1. We fix a Frobenius pair (G, H) . We denote by N the corresponding Frobenius kernel. The key result of the proof is the following proposition:

Proposition 12. Let χ be an irreducible character of H . There exists $\tilde{\chi}$ an irreducible character of G such that $\tilde{\chi}|_H = \chi$ and $\tilde{\chi}$ is constant on N .

Remark 1. It is easy to see that every central function on H extends in a unique way into a central function on G constant on N . What matters here is that this extension is a character (irreducible).

The following lemma will allow us to produce characters constant on N :

Lemma 2. Let $f \in \text{Cent}(H)$ such that $f(e) = 0$ and $\tilde{f} = \text{Ind}_H^G(f)$. Then $\tilde{f}|_H = f$ and $\tilde{f}|_N = 0$

Proof. This is immediate by the formula defining the induced function that $\tilde{f}(e) = 0 = f(e)$.

Let $g \in H \setminus \{e\}$. Notice that if $k \in G, kgk^{-1} \in H \iff g \in k^{-1}Hk \cap H \iff k \in H$ by definition of a Frobenius complement. We thus have $\tilde{f}(g) = \frac{1}{|H|} \sum_{k \in G, kgk^{-1} \in H} f(kgk^{-1}) = \frac{1}{|H|} \sum_{k \in H} f(kgk^{-1}) = \frac{1}{|H|} \cdot |H|f(g) = f(g)$ because f is central on H .

Let $g \in N \setminus \{e\}$, for $k \in G, kgk^{-1} \notin H$ (otherwise, $g \in k^{-1}Hk$, a contradiction by the definition of N), hence $\tilde{f}(g) = 0$. $\square$

Proof of proposition 12. The reader may refer to the table in remark 2 after the proof which summarizes the functions involved. We reuse the notations of proposition 12.

Let $d = \chi(e)$ be the degree of χ , $l = |G|/|H| \in \mathbb{N}$, θ_H the trivial character of H and θ_G that of G . If χ is the trivial character, it suffices to take $\tilde{\chi}$ the trivial character of G which is irreducible.

Otherwise, in order to apply lemma 2, we consider $\chi - d \cdot \theta_H \in \text{Cent}(H)$ which vanishes at e .

Then, $\tilde{\psi} := \text{Ind}_H^G(\chi - d \cdot \theta_H)$ vanishes on N and has $\chi - d \cdot \theta_H$ as its restriction to H (*). However, it is not a character (because $\tilde{\psi}(e) = 0$). In order to obtain an irreducible character, we need at least a central function of Hermitian norm 1. We therefore compute, with proposition 11,

$$\langle \tilde{\psi}, \tilde{\psi} \rangle_G = \langle \text{Ind}_H^G(\chi - d \cdot \theta_H), \tilde{\psi} \rangle_G = \langle \chi - d \cdot \theta_H, \tilde{\psi}|_H \rangle_H = \langle \chi - d \cdot \theta_H, \chi - d \cdot \theta_H \rangle_H = 1 + d^2$$

where we used the orthogonality of the irreducible characters and the fact that $\chi \neq \theta_H$. By proposition 11,

$$\langle \tilde{\psi}, \theta_G \rangle_G = \langle \text{Ind}_H^G(\chi - d \cdot \theta_H), \theta_G \rangle_G = \langle \chi - d \cdot \theta_H, \theta_{G|H} \rangle_H = \langle \chi - d \cdot \theta_H, \theta_H \rangle_H = -d$$

We are therefore led to define $\tilde{\chi} = \tilde{\psi} + d \cdot \theta_G$ which gives (where the inner products are taken in $\text{Cent}(G)$)

$$\langle \tilde{\chi}, \tilde{\chi} \rangle = \langle \tilde{\psi}, \tilde{\psi} \rangle + 2d \langle \tilde{\psi}, \theta_G \rangle + d^2 \langle \theta_G, \theta_G \rangle = 1 + d^2 - 2d^2 + d^2 = 1$$

Moreover, $\tilde{\chi}$ is clearly a linear combination of characters with coefficients in $\mathbb{Z}$. There therefore exist $n_1, \dots, n_r \in \mathbb{Z}$ such that $\tilde{\chi} = \sum_{i=1}^r n_i \chi_i$. Then $1 = \langle \tilde{\chi}, \tilde{\chi} \rangle = n_1^2 + \dots + n_r^2$ hence since the n_i are integers, only one of the n_i is nonzero and equals ± 1 . Since $\tilde{\chi}(e) = d > 0$ (cf. below), this n_i is equal to 1 and $\tilde{\chi}$ is an irreducible character.

Finally, $\tilde{\chi}$ remains constant on N (we added $d \cdot \theta_G$ to $\tilde{\psi}$ which was constant on N), and for $h \in H$, $\tilde{\chi}(h) = \tilde{\psi}(h) + d \cdot \theta_G(h) = (\chi - d \cdot \theta_H)(h) + d = \chi(h)$ then $\tilde{\chi}|_H = \chi$, which completes the proof. $\square$

Remark 2. In fact, we know all the values of $\tilde{\chi}$. Indeed, $G = N \cup \bigcup_{g \in G} gHg^{-1}$, and no element of N is conjugate to an element of $\bigcup_{g \in G} gHg^{-1} \setminus \{e\}$ (otherwise it would be conjugate to an element of $H \setminus \{e\}$). Moreover, every element of $\bigcup_{g \in G} gHg^{-1} \setminus \{e\}$ is conjugate to an element of H . Thus, if we denote by $A_1, \dots, A_m$ the conjugacy classes in G of the elements of $\bigcup_{g \in G} gHg^{-1} \setminus \{e\}$, we have $\bigcup_{g \in G} gHg^{-1} \setminus \{e\} = \sqcup_{i=1}^m A_i$ and we have $h_1 \in A_1 \cap H, \dots, h_m \in A_m \cap H$. We then have the following table (since the characters are central)

	e	A_1	...	A_m	$N \setminus \{e\}$
$\tilde{\psi}$	0	$\chi(h_1) - d$	...	$\chi(h_m) - d$	0
θ_G	1	1	...	1	1
$\tilde{\chi} = \tilde{\psi} + d \cdot \theta_G$	$d = \chi(e)$	$\chi(h_1)$	...	$\chi(h_m)$	$d = \chi(e) = \tilde{\chi}(e)$

The decomposition of a character into irreducible characters also yields the corollary:

Corollary 4. Let χ be a character of H . There exists $\tilde{\chi}$ a character of G such that $\tilde{\chi}|_H = \chi$ and $\tilde{\chi}$ is constant on N . The table above remains valid.

One final lemma will allow us to conclude.

Lemma 3. Let χ be a character of G afforded by a representation ρ . Then $\chi(g) = \chi(e)$ if and only if $g \in \ker(\rho)$

Proof. Let $d = \chi(e)$ be the degree of ρ . If $g \in G$, as seen in Example 5, $\rho(g)$ is diagonalizable with spectrum $\{\lambda_1, \dots, \lambda_d\}$ in $\mathbb{U}$. Then the triangle inequality gives $|\chi(g)| = |\lambda_1 + \dots + \lambda_d| \leq |\lambda_1| + \dots + |\lambda_d| = d = \chi(e)$ with equality if and only if $\lambda_1, \dots, \lambda_d$ are positively collinear hence if and only if they are equal (because of modulus 1). So $\lambda_1 + \dots + \lambda_d = d$ if and only if $\lambda_1 = \dots = \lambda_d = 1$ if and only if $\rho(g) = \text{id}$. $\square$

At this point, all the ingredients are in place to conclude.

Completion of the proof of theorem 1. Take ρ a faithful representation of H (for example the regular representation) and χ its character. Let $\tilde{\chi}$ be as defined in corollary 4 and $\tilde{\rho}$ the associated representation. We show that

$$N = \ker(\tilde{\rho})$$

We have, by the table of remark 2 and lemma 3, $\ker(\tilde{\rho}) = \{x \in G, \tilde{\chi}(x) = \tilde{\chi}(e)\} = N$ (where, for the last equality, we used the following fact: $\chi(h_1), \dots, \chi(h_m) \neq \chi(e)$ by lemma 3 and the faithfulness of ρ).

N is therefore a (normal) subgroup of G , which completes the proof. $\square$

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6 Appendix: useful prerequisites

Throughout this section, let $(G, \cdot)$ be a finite group.

Definition A. Let X be a set. An action of the group G on X is a homomorphism φ from G to $\mathfrak{S}(X)$. We write $g \cdot x := \varphi(g)(x)$ for $(g, x) \in G \times X$. If $x \in X$, we call the stabilizer of x the set $\text{Stab}(x) = \{g \in G, g \cdot x = x\}$. One checks that it is a subgroup of G . We say that the action is faithful if φ is injective. We say that the action is transitive if $\forall x \in X, \{g \cdot x, g \in G\} = X$.

Proposition A. Suppose that G acts on X . Let $g \in G$ and $x \in X$. $\text{Stab}(g \cdot x) = g\text{Stab}(x)g^{-1}$.

Proof. If $h \in \text{Stab}(g \cdot x)$, $h \in \text{Stab}(g \cdot x) \iff h \cdot (g \cdot x) = g \cdot x \iff (g^{-1}hg) \cdot x = x \iff g^{-1}hg \in \text{Stab}(x) \iff h \in g\text{Stab}(x)g^{-1}$ $\square$

Definition B. Let $(G, \cdot)$ be a group.

- If $g \in G$, the map $\varphi_g : x \in G \mapsto gxg^{-1} \in G$ is a group automorphism, called the conjugation automorphism, and the map $g \in G \mapsto \varphi_g \in \text{Aut}(G)$ is an action of G on itself.
- x and y in G are said to be conjugate if there exists $g \in G$ such that $x = \varphi_g(y)$ this defines an equivalence relation on G whose classes are called conjugacy classes. We define similarly the conjugates of a subset of G . If H is a subgroup of G , for every $g \in G$, $\varphi_g(H)$ is a subgroup of G (as the image of a subgroup under a morphism) of cardinality $|H|$ (because φ_g is injective).

Definition C. Let H be a subgroup of G . We say that H is "distinguished" or "normal" if

$$\forall g \in G \forall h \in H, ghg^{-1} \in H$$

Definition D. Let H be a subgroup of G . We say that $x \in G$ and $y \in G$ are "congruent modulo H " if $y^{-1}x \in H$. One checks that this defines an equivalence relation on G whose classes ("left cosets modulo H ") are the $gH, g \in G$. We use the notation $G/H = \{gH, g \in G\}$. Since $\tau_g : h \in G \mapsto gh$ is injective for $g \in G$, the elements of G/H all have cardinality $|H|$ hence, since their union is G , $|G/H| = |G|/|H|$ (which shows in particular that $|H|$ divides $|G|$: this is Lagrange's theorem).

Finally, we define the notion of Hermitian space which generalizes the notion of a Euclidean space:

Definition E. Let V be a $\mathbb{C}$ -vector space. We call a Hermitian inner product on V any map $\langle, \rangle : V^2 \rightarrow \mathbb{C}$ which is linear in the 2nd variable ($\forall (x, y, z, \lambda) \in V^3 \times \mathbb{C}, \langle x, y + \lambda z \rangle = \langle x, y \rangle + \lambda \langle x, z \rangle$), Hermitian symmetric ($\forall (x, y) \in V^2, \langle x, y \rangle = \overline{\langle y, x \rangle}$), positive ($\forall x \in V, \langle x, x \rangle \in \mathbb{R}^+$), and definite ($\forall x \in V, \langle x, x \rangle = 0 \iff x = 0$). We define as in the Euclidean case the notion of orthogonality and of orthonormal families.